

Group Project

Total Project Marks (10 Marks)

Instructions

1. Form a group of 5 members.
2. Brainstorm Embedded system project ideas collectively and evaluates them based on feasibility, impact, and educational value.
3. Your project can be in the following areas (but not limited):

Smart Home Automation	Environmental Monitoring	Smart Retail Systems
Health Monitoring	Smart Security and Surveillance	Wearable Embedded Systems
Smart Agriculture	Assistive Technology	Smart Infrastructure Monitoring
Traffic Monitoring	Transportation Safety Systems	Home Healthcare Systems
Smart City	Smart Waste Management	Disaster Management Systems
Smart Energy Management	IoT-Based Monitoring Systems	Industrial Automation

4. Detail out the requirements (functional and non-functional) of your project that will be able to solve the problems identified in **Instruction 3**.

5. **Project Final Deliverable:**

- a. Develop the project in TinkerCad for this embedded control system based on an Arduino board.
Note: You may also use the Arduino IDE and implement the project using actual Arduino hardware; however, is not compulsory.
- b. Design a complete and professionally drawn circuit for your solution.
- c. Construct an appropriate **Finite State Machine (FSM)** diagram for the system you are to develop.
- d. Develop a complete C/C++ code based on the FSM designed in **part (c)**. You may develop, test, and save your code in the Tinkercad simulator. Alternatively, you may use the Arduino IDE if you plan to build and test the project using actual Arduino hardware.
- e. Make a **PDF report** including project introduction, problem statement, scope, goals, finite state machine (FSM) diagram, Tinkercad Circuit view and Schematic View, Component list, C/C++ code.
- f. The submission of the project should be via ULearn.
- g. Deadline for submission is on **20th March 2026 (Friday), 11:59 pm (Week 10)**.